Mapping the Stellar Fossil Record of M101 with HETDEX:
Spatially Resolved Ages and Metallicities

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BACKGROUND:

Galaxies evolve through external and internal mechanisms, and these processes are encoded in the "fossil record" of a galaxy's stellar populations (Tinsley 68). To see this record, we need spatially resolved data of stellar ages and metallicities which we can only get from nearby galaxies. The Pinwheel Galaxy (M101) is a perfect target since it has spent most of its history isolated but has recently undergone an interaction with its most massive satellite companion, NGC 5474.

METHODS

1. Collected integral field spectroscopy using VIRUS at the Hobby-Eberly Telescope and combined with archival multiwavelength photometry.
2. Extracted an integrated spectrum as well as binned spatially resolved spectra.

RESULTS

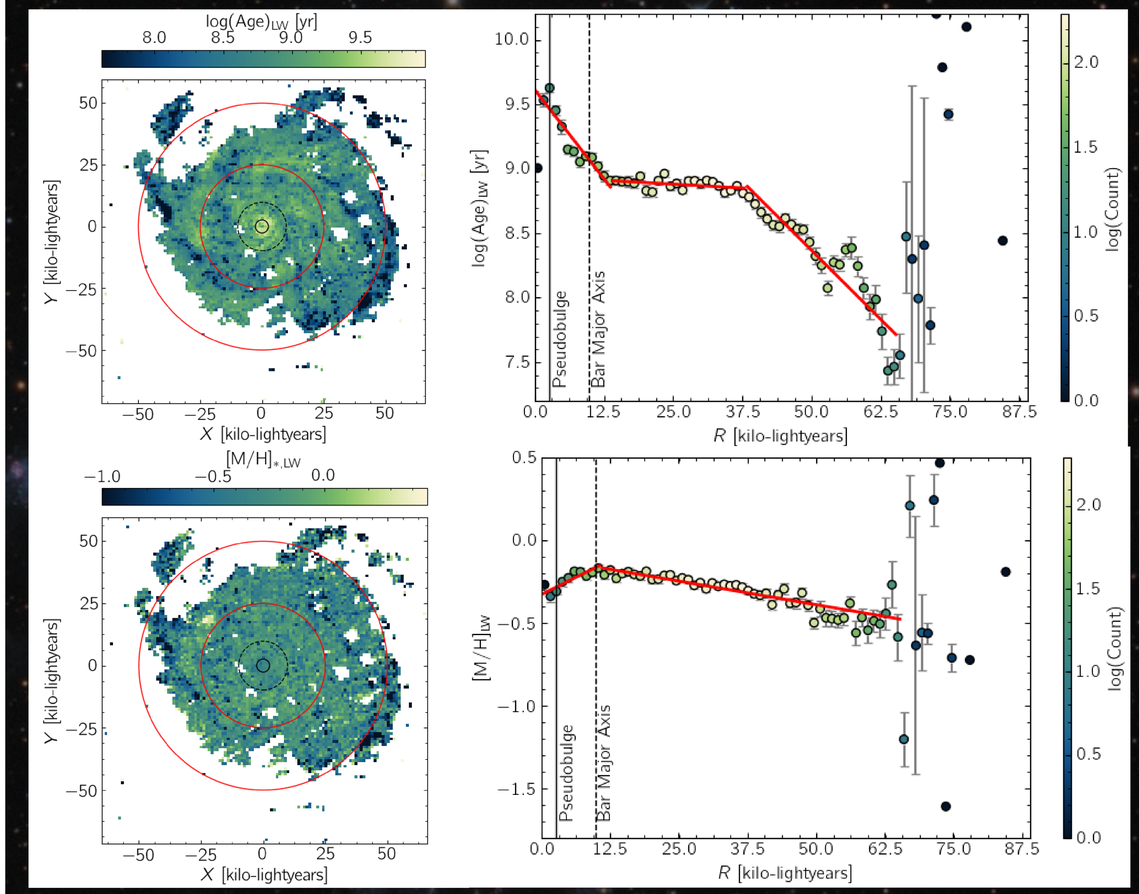
I produced the first integrated spectrum of M101 and computed globally averaged properties with pPXF (Cappellari 23) and Bagpipes (Carnall+19), finding it to be consistent with being a massive spiral galaxy, with stellar ages and a star-formation history indicating a recent burst of star formation, likely from its interaction with NGC 5474 (Linden & Mihos+22).

I constructed 2D stellar age and metallicity maps as well as gradients, finding:

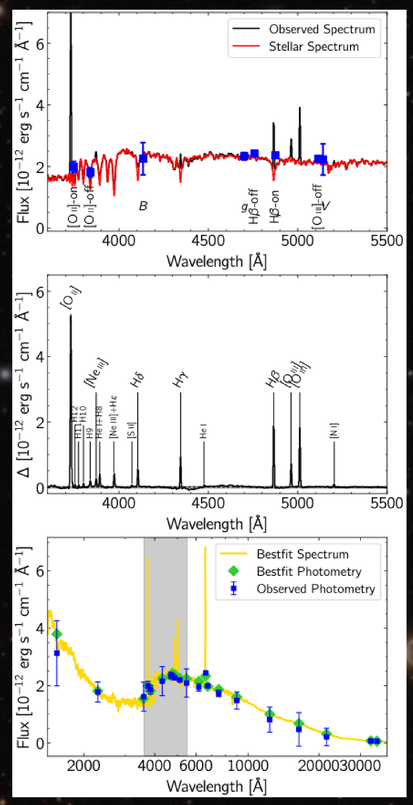
- Evidence of internal evolution in the center and inner disk (young spiral arms, low metallicity near the bulge and bar radius, etc.)
- Suggestive evidence of external effects in the outer disk at the edge of our data coverage (burst of star formation, evolving steep metallicity gradient, etc.)

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Using the first complete spectroscopic coverage of M101, we show that its stellar populations show signs of long-lived internal evolution and very recent external evolution.



The stellar ages and metallicities across M101. Maps and radial gradients for the ages (top row) and metallicities (bottom row). Red circles on the maps have radii of 25 and 50 kilo-lightyears. The solid black circle is the radius of the pseudobulge, and the dashed black circle is the radius of the bar. Broken linear fits are plotted on the right.

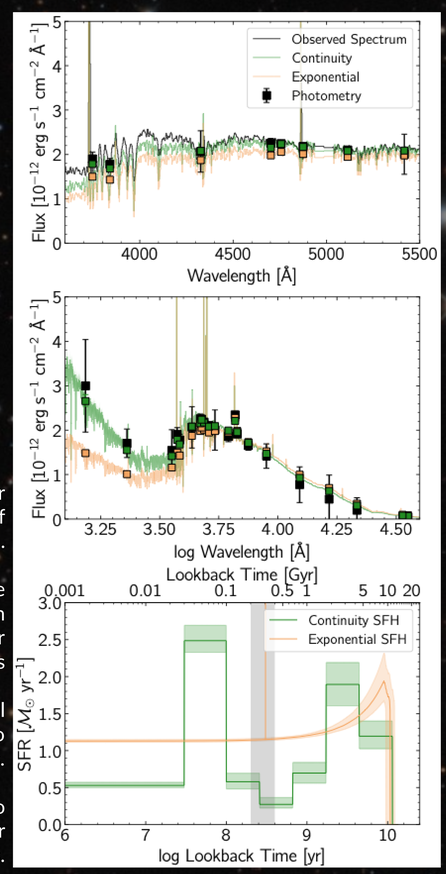


The integrated spectrum of M101.

Top panel: the observed spectrum compared to models and observed photometry.

Middle panel: strong and weak gas emission lines recovered after subtracting the stellar continuum.

Bottom panel: the full UV-IR spectrum with archival photometry.



The integrated star formation history of M101.

Top panel: the observed spectrum compared to two star formation histories

Middle panel: the full UV-IR spectrum for two star formation histories.

Bottom panel: the two recovered star formation histories.

Take a picture to **download** the **full paper**

References
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Carnall et al. 2019, MNRAS, 490, 417
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Tinsley 1968, ApJ, 151, 547